

Towards Speech ALDi: Predicting the Arabic Level of Dialectness Directly from Speech

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Abstract

Arabic Dialect Identification (ADI) aims at automatically assigning an utterance—written or spoken—to a single variety, be it Modern Standard Arabic (MSA) or a variety of Dialectal Arabic (DA). To complement ADI, the Arabic Level of Dialectness (ALDi) has been proposed to model the different dialect levels that exist between MSA and Highly Dialectal text. However, current ALDi estimation models operate on text inputs only. This paper presents the first step towards extending ALDi to the speech modality. First, we use the automatically estimated ALDi score of a spoken utterance’s transcript as the silver-standard target score. We then compare the performance of two approaches at automatically predicting these silver-standard ALDi scores from spoken utterances: (1) a cascaded approach where automatic speech recognition is used to generate automatic transcripts that are then fed to a text-based ALDi estimation model, and (2) a direct ALDi estimation from speech. We use SADA—a dataset of 435 hours of manually transcribed Arabic speech—as the fine-tuning source data, and evaluate the two approaches on three datasets that cover different varieties of Arabic. Our results indicate that estimating silver-standard ALDi scores directly from speech is on par or slightly worse than the cascaded approach for in-domain data. However, the direct approach generalizes better, especially for dialects where the fine-tuned ASR backbone of the cascaded approach has relatively large word-error-rates. This paper paves the way towards predicting ALDi scores directly from speech inputs.¹

1 Introduction

Arabic is known for its diverse set of dialectal varieties, in addition to a standard variety that is perceived to be shared among the Arabic speakers, known as Modern Standard Arabic (MSA)

(Habash, 2010). MSA is the variety used in education, news broadcasts, and official written communication. In spontaneous everyday communications, Arabic speakers rely on regional dialects, that have variation in phonology, morphology, syntax, and lexical items (Holes, 2006).

Arabic Dialect Identification (ADI) is typically employed to identify the variety of the different samples of a dataset. Given an input written or spoken utterance, ADI aims at assigning the utterance to a single Arabic variety. However, this formulation adapts a binarized distinction between MSA and the other dialectal varieties of Arabic, marginalising the different dialect levels that exist in between (Badawi, 1973). To mitigate this, Keleg et al. (2023) introduced the Arabic Level of Dialectness (ALDi), a continuous score in the range [0, 1], which models the different levels between MSA (ALDi=0) and highly dialectal Arabic (ALDi=1).² ALDi proved to be a useful tool for multiple applications, such as better routing of samples to annotators (Keleg et al., 2024), evaluating the LLMs’ dialectal capabilities (Robinson et al., 2025), and building better recommendation systems (Alshabanah and Annamaram, 2025).

However, ALDi operates on text inputs. Hence, there is a need for adapting ALDi to the speech modality, as explicitly acknowledged by Sullivan et al. (2026). Applying text-based ALDi models to speech inputs requires first transcribing them into text, where transcription errors can reduce the accuracy of the estimated ALDi scores. Additionally, some phonological variations in speech are marginalized in text. For instance, the word قلب (heart) is a shared term between MSA and many dialectal varieties. It is typically written according to the MSA standardized spelling قلب /qlb/, even when it could be pronounced as /glb/ or /ʔlb/ in dif-

¹Our codebase can be accessed through [this link](#).

²Arabic Generality Score was later proposed as an orthogonal dimension to ALDi, measuring how widely a word is used in different Arabic dialects (Sha’ban and Habash, 2025).

ferent Arabic varieties, for which a phonologically faithful spelling is **جلب** and **ألب** respectively.

In this paper, we take the first step towards adapting ALDi estimation to speech inputs. For a given speech input, we feed its manual transcript to a text-based ALDi estimation model to generate the silver-standard ALDi score. We then compare the performance of two approaches: (1) a cascaded approach of first generating an automatic transcript that is fed to the text-based ALDi estimation model, and (2) a speech-based model that directly estimates the ALDi score. We use two large pre-trained models in our experiments: Whisper-medium (Radford et al., 2023) and MMS-1B (Pratap et al., 2024), and employ the SADA corpus of Saudi Arabic speech (Alharbi et al., 2024) as the fine-tuning dataset, using automatically estimated silver-standard ALDi scores as target labels. We evaluate the models on three different datasets: in-domain Saudi speech, broadcast speech, and dialectal speech.

We believe that this paper paves the way towards establishing speech ALDi as a feasible extension to the text-based one. However, future work should study how to define ALDi scores for speech inputs. For instance, speech-based ALDi scores might consider accounting for the deviations in the pronunciation of MSA-shared lexical terms in some dialects, whereby these deviations are typically normalized in text.

2 Silver-Standard ALDi scores of Speech Utterances

The ALDi score of a piece of text is operationalised as a continuous score in the range $[0, 1]$, where 0 represents pure MSA text, 1 represents highly dialectal text, and the intermediate scores represent the dialect levels that exist between MSA and highly dialectal text (Keleg et al., 2023). To acquire the gold-standard ALDi scores of the AOC-ALDi dataset (Keleg et al., 2023), three annotators rated pieces of text on a discrete four-level scale: MSA ($ALDi=0$), Little Dialectal ($ALDi=\frac{1}{3}$), Mixed ($ALDi=\frac{2}{3}$), and Mostly Dialectal ($ALDi=1$). The target ALDi score for a piece of text is the mean of its three ALDi ratings. The AOC-ALDi was used to fine-tune a regression head on top of a pretrained Arabic BERT-based model, resulting in the *Sentence-ALDi* model that can automatically estimate the ALDi scores of pieces of text.³

³<https://huggingface.co/AMR-KELEG/Sentence-ALDi>

Acquiring ALDi scores for spoken utterances poses two challenges. First, the different dialect levels used to annotate text utterances need to account for the phonological variations that typically get normalized in text, but exist in speech. Second, manual annotation of a large number of spoken utterances needs to be done, which is a non-trivial task, especially since estimating ALDi scores requires having annotators speaking different Arabic dialects, as it was found that perception of the ALDi scores can be influenced by the annotators’ native dialects (Keleg et al., 2025).

In this paper, we approximate the ALDi score of a spoken utterance as the ALDi score of its corresponding transcript. For a spoken utterance (U_{spoken}) with a corresponding transcript (U_{text}), we assume that $ALDi(U_{\text{spoken}})$ is equal to $ALDi(U_{\text{text}})$. Additionally, we approximate $ALDi(U_{\text{text}})$ with the automatically estimated ALDi scores of the transcript $ALDi(U_{\text{text}})$ using the *Sentence-ALDi* model, forming the approximate silver-standard ALDi target scores for the spoken utterance $ALDi(U_{\text{spoken}})$.

3 Experimental Details

In this section, we describe the datasets and the models we used to estimate the ALDi scores of spoken utterances.

3.1 Fine-tuning Dataset

SADA (Alharbi et al., 2024) is a dataset of 435 hours of manually transcribed Arabic speech, sourced from 57 television shows provided by the Saudi Broadcasting Authority. The dataset mostly covers Saudi dialects, but it also includes other dialects such as Yemeni, Egyptian, and Levantine. We use SADA’s training set (417.6 hours) to fine-tune our models, as elaborated next.

3.2 Modeling Approaches

Approach #1 - Cascaded ASR-ALDi Estimation

In this setup, an ASR model is used to automatically transcribe a spoken utterance (U_{spoken}) to ($U_{\text{text}} - \text{automatic}$). Afterwards, the automatic transcription ($U_{\text{text}} - \text{automatic}$) is fed to the text-based *Sentence-ALDi* model to estimate the ALDi score.

For the underlying ASR model, we use the pre-trained *Whisper-medium* model (Radford et al., 2023). We also study the impact of fine-tuning the *Whisper-medium* model on the speech-transcript pairs of SADA’s training set, motivated by the strong accuracy that fine-tuned Whisper models achieve on ADI benchmarks (Elleuch et al., 2025).

Approach #2 - Direct ALDi Estimation from Speech

In this setup, we fine-tune speech models to automatically estimate the ALDi scores directly from speech inputs. We use a speech-based encoder to generate a continuous representation for the input speech’s frames. We then perform mean pooling of these representations into a single vector that is fed to a linear projection layer to predict a scalar ALDi score. We experiment with two speech encoders: *Whisper-medium*’s encoder,⁴ and *MMS-1B* (Pratap et al., 2024), a wav2vec 2.0-based model, pre-trained on over 1,400 languages and operating on raw waveforms. Refer to Appendix §A for further discussion of the training setup and the employed hyperparameters.

3.3 Test Datasets

To evaluate the fine-tuned ALDi estimation models, we employ three different testing sets:

SADA_(test set) (Alharbi et al., 2024) consists of 10.8 hours of manually transcribed speech, serving as the in-domain test set of our ALDi estimation models. The mean value of the silver-standard ALDi scores of this dataset is 0.62, indicating a significant representation of dialectal speech utterances compared to MSA.

MediaSpeech (Kolobov et al., 2021) provides 10 hours of manually transcribed Arabic speech, extracted from the YouTube channels of three news broadcasting channels: Al-Arabiya, France 24 Arabic, and BBC News Arabic. This dataset is expected to be dominated by MSA speech (typically used in news broadcasts), as indicated by the mean of the silver-standard ALDi scores of this dataset, 0.15. This dataset is used to test whether the models assign low ALDi scores to MSA utterances, which were previously estimated to only represent 1.8% of SADA’s training set (Alharbi et al., 2024).

Casablanca (Talafha et al., 2024) is a multi-dialectal speech dataset, focusing on eight varieties of Arabic dialects, representing four regional dialects: Levant (Jordan, Palestine), Gulf (UAE, Yemen), Nile Basin (Egypt), and Maghreb (Algeria, Morocco, Mauritania). We use the Casablanca’s test set for out-of-domain evaluation, whereby each country has a total of approximately 1 hour (i.e.,

⁴We also tested fine-tuning Whisper-small and Whisper-large-v3 on a sample of the fine-tuning dataset, finding that Whisper-medium outperforms them.

Test Set	(1) Cascaded Models		(2) Direct Models	
	Whisper _{med}		Whisper _{med}	MMS-1B
	Zero-shot	Fine-tuned		
SADA _(test set)	0.396	0.184	0.191	0.204
MediaSpeech	0.177	0.112	0.114	0.122
Casablanca	0.393	0.210	0.199	0.206
Mean RMSE	0.32±0.13	0.17±0.05	0.17±0.05	0.18±0.05

Table 1: The RMSE ($\downarrow$) of the estimated ALDi scores, for the three test sets. The RMSE of the best-performing model on each test set is shown in **bold**.

Note: the text-based ALDi model has a similar overall RMSE (0.18) on the test set used in Keleg et al. (2023).

8 hours in total). The mean of the silver-standard ALDi scores of this test set is 0.76, indicating a skew towards highly dialectal speech.

4 Results

Table 1 reports the performance of the two approaches, as measured by the RMSE between the silver-standard target ALDi scores and the ALDi scores estimated by each model.

The zero-shot setup of the cascaded model performs substantially worse than the other approaches, hinting that a pre-trained ASR model without domain adaptation is insufficient for reliable ALDi estimation. For the direct models, Whisper-medium performs slightly better than MMS-1B across the three test sets.

Overall, the fine-tuned cascaded model and the direct Whisper-based model exhibit comparable performance. The cascaded approach achieves lower RMSE values on SADA-test (0.184) and MediaSpeech (0.112), marginally outperforming direct Whisper-medium (0.191 and 0.114, respectively). However, direct Whisper-medium performs better on Casablanca, achieving an RMSE of 0.199 compared with 0.210 for the cascaded model.⁵

Dialect-Level Performance on Casablanca To further situate the results, we contrast the results of the fine-tuned cascaded model and the direct Whisper-based models on each dialect split of Casablanca’s test set, as in Table 2. The significance of the difference in the performance of the two models is assessed using a paired two-tailed *t*-test on per-utterance squared error.

⁵We analyse the impact of the fine-tuning data size on the model’s performance in Appendix §B.

Region	Dialect	N	WER	(1) Cascaded Fine-tuned	(2) Direct Whisper _{med}
Levant	Jordan	848	0.414	0.126**	0.160
	Palestine	667	0.493	0.169	0.180
Gulf	UAE	813	0.434	0.158	0.164
	Yemen	803	0.649	0.197	0.197
Nile	Egypt	846	0.663	0.196*	0.216
Maghreb	Algeria	844	0.845	0.231	0.218
	Morocco	1,045	0.832	0.238	0.192**
	Mauritania	953	0.912	0.291	0.239**
All Dialects		6,819	0.670	0.210	0.199**

Table 2: RMSE ($\downarrow$) by dialect on the Casablanca test set, with mean WER of the fine-tuned cascaded model’s ASR transcripts. Best RMSE per dialect is shown in **bold**. * $p < 0.05$, ** $p < 0.001$, according to a paired t -test between the scores of both approaches.

A clear regional pattern emerges: the fine-tuned cascaded model achieves lower RMSEs on the Gulf, Levantine, and Egyptian dialects, whereas the direct model achieves lower RMSEs on the Maghrebi dialects. The direct model’s advantage is statistically significant for Morocco and Mauritania ($p < 0.001$) and for the full Casablanca set ($p < 0.001$). Conversely, the fine-tuned cascaded model’s advantage is significant for Jordan ($p < 0.001$) and Egypt ($p < 0.05$), while the remaining dialects show no significant difference.

This pattern could be a result of the unbalanced representation of Arabic varieties in SADA’s training split, where the Maghrebi dialects are not well-represented compared to the other dialects, which undermines the quality of the transcripts generated by the cascaded ASR-ALDi approach. The direct model is less vulnerable to this limitation, as it directly estimates the ALDi score without relying on an intermediate automatically generated transcript.

Qualitative Error Analysis On inspecting the transcripts of the cascaded model for the two dialects with the Maghrebi dialects of Casablanca, we found that the ALDi score of a sample with a high WER does not always deviate from the silver-standard ALDi score. While a high-WER transcript is not faithful to the underlying speech utterance, the corresponding ALDi score of this transcript will depend on the level of dialectness of the erroneous transcript. For instance, the two examples in Table 3 have inaccurate transcripts with high WERs (0.89 and 1.17), and high silver-standard ALDi scores (0.98 and 0.87). However, the ALDi scores estimated by the cascaded approach diverge

	Example 1	Example 2
Gold Transcript	نورمال داكشي لي شتي تا انا راه كا نفرح ملي كا ن...	ما عندك مناش تخافي صافي مشا
Whisper’s Transcript	الله مالكشي يشتهي، بس أنا كنت فرح منك نشوف الخط...	عندكم نشغل في صفي إن شاء الله.
WER	0.89	1.17
ALDi_{Target}	0.98	0.87
ALDi_{(1) Cascaded}	0.92	0.20
ALDi_{(2) Direct}	0.80	0.73

Table 3: Two qualitative examples from Casablanca’s Maghrebi samples, with high WERs. However, the impact of the erroneous transcript on the ALDi score estimated using the cascaded approach depends on whether the ASR model wrongly decoded MSA terms in the transcript or dialectal ones. The direct ALDi estimation approach is more robust than the cascaded approach in these cases.

for both cases, whereby it is high for the first (0.98) and low for the second (0.20). For the first example, the erroneous transcript still uses lots of dialectal terms (e.g., **مالكشي**, **بس**, **نشوف**), which are not part of the gold standard transcript. However, these dialectal terms are still enough to make the ALDi score of the transcript high. Conversely, the second example’s transcript mostly uses MSA terms (e.g., **إن شاء الله**, **عندكم**). Hence, the estimated ALDi score of the transcript is wrong, deviating from the silver-standard ALDi score.

5 Genre-based Extrinsic Evaluation

To further study the usability of the direct model, we evaluated it on 1,791 fifteen-second clips drawn from 49 publicly available YouTube sources (Appendix C), spanning six genres: broadcast news, religious sermons, drama, talk shows, podcasts, and street interviews, covering Gulf, Maghrebi, Egyptian, and Levantine varieties. Sources were selected to be predominantly conversational. We downloaded the first 10 minutes of each video.⁶ Then, the first and last 30 seconds of each video were removed to heuristically exclude introductory and concluding segments. The remaining audio was cut into fixed 15-second windows without sentence-boundary alignment. We use box plots to visualize the predicted ALDi scores for all clips of each genre, as per Figure 1. The model clearly separates formal from conversational speech: news scores the lowest (mean ALDi 0.042), religious speech next (0.209), while all conversational genres cluster between 0.72 and 0.80.

⁶Note that some videos are shorter than 10 minutes.

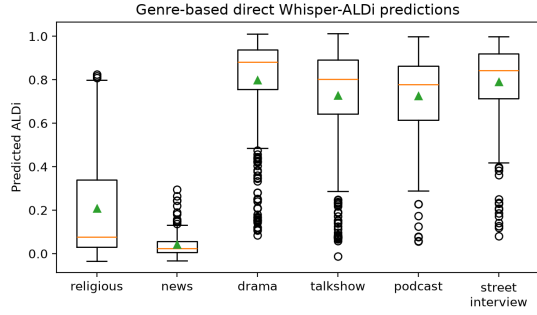


Figure 1: Distribution of predicted ALDi scores from the direct model based on Whisper-medium, split by genre. News and religious speech sit well below the conversational genres, which overlap substantially with one another.

6 Conclusion and Future Work

Fine-tuning pre-trained speech encoders on ALDi-labelled Arabic audio shows that the speech signal contains enough information to recover a useful approximation of text-based ALDi across in-domain, out-of-domain, and MSA data. A fine-tuned ASR pipeline remains the strongest overall system on SADA-test and MediaSpeech, but the direct model is competitive overall, achieving lower RMSE on Casablanca, and is more robust on Maghrebi dialects, where ASR quality degrades most. Direct speech ALDi is not a weaker substitute for the ASR-based pipeline, but a complementary approach that remains effective when transcription is unreliable.

These findings should be read as an initial step rather than a solved problem. A priority for future work is designing annotation guidelines for speech-native ALDi that explicitly account for speech-specific variables such as phonetic variation and prosody, rather than continuing to rely on transcript-derived targets, so that speech ALDi can be evaluated as a speech-native task. We see this work as a step towards establishing speech ALDi as a feasible complement to the text-based framework.

Limitations

Several limitations constrain how strongly these findings are interpreted. The training labels are not speech-native human ALDi annotations, but are silver-standard labels automatically estimated using a text-based ALDi model from manual transcripts. Hence, the direct model learns to approximate a text-based target and may inherit any biases introduced during transcription. All training data comes from the transcribed portion of SADA,

heavily centred on Saudi speech and mid-to-high ALDi values, likely contributing to weaker transfer to unseen dialects. Lastly, we only relied on simple mean pooling for the direct ALDi estimation approach, and have not experimented with other aggregation techniques.

Ethical Considerations

We used AI writing assistance within the scope described in the ACL Policy on Publication Ethics, and manually validated the accuracy and validity of its outputs.

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References

- Sadeen Alharbi, Areeb Alowisheq, Zoltán Tüske, Kareem Darwish, Abdullah Alrajeh, Abdulmajeed Alrowithi, Aljawharah Bin Tamran, Asma Ibrahim, Raghad Aloraini, Raneem Alnajim, Ranya Alkahtani, Renad Almuasaad, Sara Alrasheed, Shaykhah Alsubaie, and Yaser Alonaizan. 2024. [SADA: Saudi audio dataset for Arabic](#). In *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 10286–10290. IEEE.
- Abdulla Alshabanah and Murali Annavaram. 2025. [On using Arabic language dialects in recommendation systems](#). In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 2178–2186, Albuquerque, New Mexico. Association for Computational Linguistics.
- As-Said Muhámmad Badawi. 1973. *Mustawayat al-arabiyya al-muasira fi Misr*. Dar al-maarif.
- Haroun Elleuch, Salima Mdhaffar, Yannick Estève, and Fethi Bougares. 2025. [ADI-20: Arabic dialect identification dataset and models](#). In *Interspeech 2025*, pages 2775–2779. ISCA.
- Nizar Y Habash. 2010. *Introduction to Arabic natural language processing*. Morgan & Claypool Publishers.
- Clive Holes. 2006. [The Arabic dialects of Arabia](#). *Proceedings of the Seminar for Arabian Studies*, 36:25–34.
- Amr Keleg, Sharon Goldwater, and Walid Magdy. 2023. [ALDi: Quantifying the Arabic level of dialectness of text](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 10597–10611, Singapore. Association for Computational Linguistics.

- Amr Keleg, Sharon Goldwater, and Walid Magdy. 2025. [Revisiting common assumptions about Arabic dialects in NLP](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3309–3327, Vienna, Austria. Association for Computational Linguistics.
- Amr Keleg, Walid Magdy, and Sharon Goldwater. 2024. [Estimating the level of dialectness predicts inter-annotator agreement in multi-dialect Arabic datasets](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 778–789, Bangkok, Thailand. Association for Computational Linguistics.
- Rostislav Kolobov, Olga Okhupkina, Olga Omelchishina, Andrey Platunov, Roman Bedyakin, Vyacheslav Moshkin, Dmitry Menshikov, and Nikolay Mikhaylovskiy. 2021. [MediaSpeech: Multilanguage ASR benchmark and dataset](#). *Preprint*, arXiv:2103.16193.
- Ilya Loshchilov and Frank Hutter. 2019. [Decoupled weight decay regularization](#). In *International Conference on Learning Representations*.
- Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, Maryam Fazel-Zarandi, Alexei Baevski, Yossi Adi, Xiaohui Zhang, Wei-Ning Hsu, Alexis Conneau, and Michael Auli. 2024. [Scaling speech technology to 1,000+ languages](#). *Journal of Machine Learning Research*, 25(1):1–52.
- Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. [Robust speech recognition via large-scale weak supervision](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 28492–28518. PMLR.
- Nathaniel Romney Robinson, Shahd Abdelmoneim, Kelly Marchisio, and Sebastian Ruder. 2025. [AL-QASIDA: Analyzing LLM quality and accuracy systematically in dialectal Arabic](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 22048–22065, Vienna, Austria. Association for Computational Linguistics.
- Sanad Sha’ban and Nizar Habash. 2025. [The Arabic generality score: Another dimension of modeling Arabic dialectness](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 30002–30013, Suzhou, China. Association for Computational Linguistics.
- Peter Sullivan, AbdelRahim Elmadany, Alcides Alcoba Inciarte, and Muhammad Abdul-Mageed. 2026. [Arab voices: Mapping standard and dialectal Arabic speech technology](#). *Preprint*, arXiv:2601.13319.
- Bashar Talafha, Karima Kadaoui, Samar Mohamed Magdy, Mariem Habiboullah, Chafei Mohamed Chafei, Ahmed Oumar El-Shangiti, Hiba Zayed, Mohamedou Cheikh Tourad, Rahaf Alhamouri, Rwa
- Assi, Aisha Alraeesi, Hour Mohamed, Fakhraddin Alwajih, Abdelrahman Mohamed, Abdellah El Mekki, El Moatez Billah Nagoudi, Benelhadj Djelloul Mama Saadia, Hamzah A. Alsayadi, Walid Al-Dhabyani, and 8 others. 2024. [Casablanca: Data and models for multidialectal Arabic speech recognition](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21745–21758, Miami, Florida, USA. Association for Computational Linguistics.

A Training Details

Models are trained with AdamW (Loshchilov and Hutter, 2019) using a cosine learning rate schedule with linear warmup over the first 10% of steps. The encoder and regression head are assigned separate learning rates to reduce the risk of catastrophic forgetting (5×10^{-6} and 2×10^{-6} for the Whisper-medium and MMS-1B encoders respectively, 2×10^{-4} for both regression heads), with gradient clipping at a maximum norm of 1.0. MMS-1B is trained with mixed precision (FP16) and gradient accumulation due to its larger memory footprint. All experiments were run on NVIDIA Quadro RTX 8000 GPUs for up to 10 epochs, and the checkpoint with the best validation performance is kept for evaluation.

B Effect of Training Data Volume

Performance improves consistently as more training data becomes available, with the largest gains at smaller scales (Figure B1). Whisper-medium’s RMSE on SADA-test falls from 0.275 at 1 hour to 0.241 at 30 hours and 0.191 at 427 hours. MMS-1B starts from a considerably higher error (e.g. 0.379 on Casablanca at 1 hour) but the gap narrows with more data; Whisper-medium is more data-efficient in the low-resource setting.

C YouTube Sources Used in Extrinsic Evaluation

The 49 YouTube sources used in the genre-based validation experiment are listed in Tables C1 and C2. Sources were selected from different genres, covering Gulf, Egyptian, Levantine, and Maghrebi varieties where possible. Three videos were subsequently removed from YouTube by their uploaders and are no longer publicly accessible; their URLs are retained for the record.

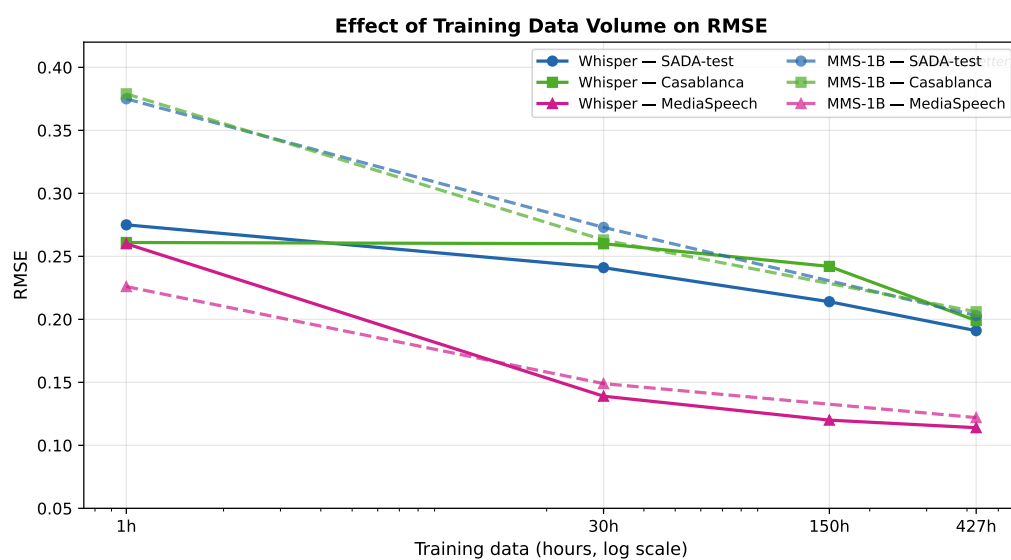


Figure B1: RMSE of Whisper-medium and MMS-1B on SADA-test as a function of training data volume. Error decreases sharply at first, with diminishing returns as data grows.

Genre	Variety	Title	URL
<i>News (4 sources)</i>			
News	MSA	خلاف مؤسسات إسلامية في فرنسا حول تحديد أول أيام رمضان	youtu.be/TXmeVPZvKq8
News	MSA	كيف تحولت تجربة خطيرة إلى أقوى سلاح تملكه أمريكا؟	youtu.be/mRP5nzHRFvQ
News	MSA	كيف ينظر الأمريكيون إلى حرب بلادهم مع إيران؟	youtu.be/rjNanL_Qfe4
News	MSA	واشنطن وطهران في مواجهة مفتوحة في مضيق هرمز	youtu.be/O_sq3kuVx9I
<i>Religious (3 sources)</i>			
Religious	CA/MSA (Mauritania)	Kindness to parents (Shaykh Muhammad Mukhtar Al-Shinqiti)	youtu.be/NBEmiMVt_eA
Religious	MSA/Lebanese	A Gospel-Centered Church, Arabic sermon (Pastor Joseph Najem)	youtu.be/2Kp_Te00KpA
Religious	CA/MSA (Kuwait)	قصة مريم بنت عمران ض نبيل العوضي	youtu.be/tXdOF6zn0sQ
<i>Drama (9 sources)</i>			
Drama	Syriab	ليالي روكسي - الحلقة ١٠	youtu.be/Sav14tXOetM
Drama	Syrian	مسلسل اليتيم - الحلقة ٣٢	youtu.be/BhKkurOzfUE
Drama	Kuwaiti	مسلسل حال منابر - الحلقة ١	youtu.be/9LHd93WxS8
Drama	Saudi	طاش ما طاش - حلقة الأيتام	youtu.be/q211ZnAVRuM
Drama	Egyptian	مسلسل الشهد والدموع - الحلقة ٤	youtu.be/CVdgG9EeR8A
Drama	Moroccan	[Video unavailable]	youtu.be/5eVakUcuZBA
Drama	Egyptian	مسلسل البيت الكبير - الحلقة ١	youtu.be/FzOKvECNWuI
Drama	Lebanese	عشق النساء - الحلقة ١	youtu.be/DTQmG0xAg
Drama	Moroccan	[Video unavailable]	youtu.be/eNU5XEbI81A
<i>Talk Show (12 sources)</i>			
Talk show	Saudi	وش صاير في رمضان - الموسم الثالث - الحلقة ٥٧	youtu.be/BfnJ-be4a8Y
Talk show	Lebanese	أحلى صباح - الخميس ٢١ آذار ٢٠٢٢	youtu.be/1TQrCEji3kY
Talk show	Jordanian	سر السعادة الزوجية - صبح صح	youtu.be/v00Bx-YTpaU
Talk show	Moroccan	صباحيات - الجمعة ٣١ مارس ٢٠٢٢	youtu.be/DWtbJv5VK9U
Talk show	Egyptian	أهلا بالستات - ٠٢ رمضان	youtu.be/7wZ43577uoM
Talk show	Egyptian	أهلا بالستات - ٢١ رمضان	youtu.be/tzgs8g9jzQM
Talk show	Egyptian	رامز جلال في معركة غير متوقعة مع يارا السكري	youtu.be/DtV2W7PRa80
Talk show	UAE	تميمة النيسر - هيئة تنمية المجتمع دبي	youtu.be/q9I77cq9Fys
Talk show	Saudi	هذا مكانك - الموسم الثالث - الحلقة ٣	youtu.be/1gHHKm6wYZE
Talk show	Moroccan	شاف فدارو - الأحد ٥١ مارس ٢٠٢٢	youtu.be/RXo1bNVJRvs
Talk show	Moroccan	رشيد شو - الجمعة ٣١ مارس ٢٠٢٢	youtu.be/SCBxthgLHz0
Talk show	Lebanese	كواليس التصوير - المحافظة ٥١	youtu.be/DJQjEiSt-j5U

Table C1: YouTube sources: News, Religious, Drama, and Talk Show (Part 1 of 2).

Genre	Variety	Title	URL
<i>Podcast (9 sources)</i>			
Podcast	Saudi	كيف تضمن الشركات استدامتها ض بودكاست سواف بزنس	youtu.be/HZKdvcoyzwI
Podcast	Moroccan	State of Dev in Morocco 2025	youtu.be/iigsuJ4YP44
Podcast	Lebanese	12 The Podcast, Ep. 17: Why Lebanese TV is dying	youtu.be/D6XyQbE7xTM
Podcast	Kuwaiti	أشهر القضايا القانونية في الكويت ض بودكاست غمد	youtu.be/Q3B1MCSbNm0
Podcast	Saudi	بودكاست خارج المشهد ض محمد الزهراني	youtu.be/HzbqbqiRPd8
Podcast	Egyptian	اللواء وليد السيسي - أسئلة حرجة مع مجدي الجلال	youtu.be/YwZh3ddj6w8
Podcast	Egyptian	الحلقة ١١ نيفين الجندي ض ازاي تلاقى نفسك	youtu.be/w0JHMyGkcf8
Podcast	Levantine	Beyond Limits: Michael Haddad	youtu.be/0ngmeMkX8FQ
Podcast	Moroccan	حكومة أخنوش والإعلام: كيف تُدار السياسة المغربية؟	youtu.be/fXukCc1aA6g
<i>Street Interview (12 sources)</i>			
Street	Saudi	مقلب اللغات مع الناس - ديقو صرنكس	youtu.be/2Gdm3zAJLkC
Street	Saudi	ضحكني ولك ٠٠٥ ريال - مقابلات الشارع	youtu.be/Aewctu7XCbY
Street	Egyptian	We Asked Egyptians What They Were Doing (Easy Egyptian Arabic)	youtu.be/Zsz0ou4VX2g
Street	Jordanian	Interviews with locals in Amman, Jordan	youtu.be/kTxGFrljLzM
Street	Moroccan	ما هي أركان الإسلام؟ أجوبة صادمة من الشارع المغربي	youtu.be/k9-vYG6I18o
Street	Moroccan	سورة المسد - أجوبة غير متوقعة من المغاربة	youtu.be/ZoEmKRuIE_4
Street	Egyptian	مواصفات فتاة أحلامك هتموت من الضحك	youtu.be/BNB9NzTt6aI
Street	Egyptian	اكسينت المذبعة بتاعتنا تحفة	youtu.be/WwazOTabYF4
Street	Jordanian	مرتاح بحياتك الزوجية؟ من مادبا مع بلال العجارمة	youtu.be/jjD976SmVSA
Street	Lebanese	[Video unavailable]	youtu.be/SDAXKba2DRs
Street	Saudi	سألتهم ايش الدولة العربية اللي يطمنون الزواج منها	youtu.be/0_04hE7TQ9U
Street	Saudi	رأي السعوديين في التغييرات والانفتاح في بلدهم	youtu.be/LzRcR4PxJR4

Table C2: YouTube sources: Podcast and Street Interview (Part 2 of 2).